

ISEN/DAEN 427 Lecture 1

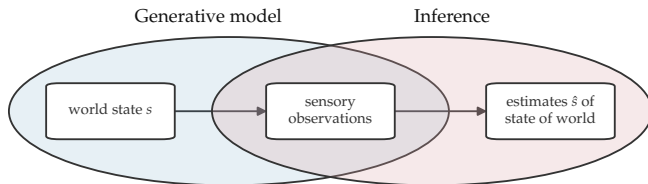
Alexander Abuabara

Perception as Inference

BM Ch. 1, 2

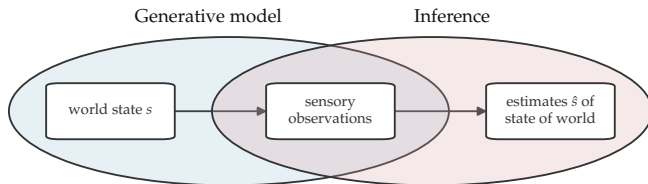


Perception as Inference



Representation of the World and Inference

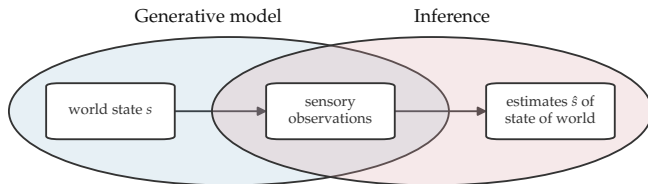
Perception as Inference



Representation of the World and Inference

- Using sensory observations to determine the “state” of the world.
- Given prior $\mathbb{P}(s)$ and data $\mathcal{D}$ with likelihood $\mathbb{P}(\mathcal{D}|s)$, the posterior distribution is: $\mathbb{P}(s|\mathcal{D}) = \frac{\mathbb{P}(\mathcal{D}|s)\mathbb{P}(s)}{\mathbb{P}(\mathcal{D})}$

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- Define *preferences* over actions $a \in A$ given rewards $r(s, a)$.
- Expected utility $a \succeq a'$ if $\mathbb{E}_{s \sim \mathbb{P}(\cdot|\mathcal{D})}[U(r(s, a))] \geq \mathbb{E}_{s \sim \mathbb{P}(\cdot|\mathcal{D})}[U(r(s, a'))]$

Perception as Inference

We define a preference relation over actions using *expected utility*.

Given two actions a and a' , we say that action a is (weakly) preferred to action a' if the expected utility of a is at least as large as that of a' :

$$a \succeq a' \quad \text{if and only if} \quad \mathbb{E}_{s \sim \mathbb{P}(\cdot | \mathcal{D})} [U(r(s, a))] \geq \mathbb{E}_{s \sim \mathbb{P}(\cdot | \mathcal{D})} [U(r(s, a'))].$$

- a, a' are actions (choices, decisions).
- s denotes the state of the world.
- $\mathbb{P}(\cdot | \mathcal{D})$ is the probability distribution over states, given data $\mathcal{D}$.
- $r(s, a)$ is the reward obtained by taking action a in state s .
- $U(\cdot)$ is the utility function, mapping rewards to subjective values.

In words: *An action is preferred whenever its expected utility, averaged over uncertain states, is no less than the expected utility of any alternative.*

Perception as Inference

The notation $s \sim \mathbb{P}$ is read as:

“the random variable s is drawn from the probability distribution $\mathbb{P}$.”

Formally, it indicates that the random variable s follows (or is distributed according to) the probability measure $\mathbb{P}$.

For example, when we write $\mathbb{E}_{s \sim \mathbb{P}}[f(s)]$, we mean the expected value of $f(s)$ when s is distributed according to $\mathbb{P}$.

In explicit terms:

$$\mathbb{E}_{s \sim \mathbb{P}}[f(s)] = \begin{cases} \int f(s) \mathbb{P}(ds) & \text{if } s \text{ is continuous,} \\ \sum_s f(s) \mathbb{P}(s) & \text{if } s \text{ is discrete.} \end{cases}$$

Example 1: Dice Roll

Let s represent the outcome of a fair six-sided die roll, so $s \in \{1, 2, 3, 4, 5, 6\}$.

The probability distribution is $\mathbb{P}(s) = \frac{1}{6}$ for each s .

Let $f(s) = s^2$ (You could pick almost any other function here, s^2 make the calculation simple, illustrative).

Then

$$\mathbb{E}_{s \sim \mathbb{P}}[f(s)] = \sum_{s=1}^6 f(s) \mathbb{P}(s) = \frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2}{6} = \frac{91}{6} \approx 15.17.$$

Example 2: Normal Distribution

Let $s \sim \mathcal{N}(0, 1)$, i.e., s is a standard normal random variable with density

$$p(s) = \frac{1}{\sqrt{2\pi}} e^{-s^2/2}.$$

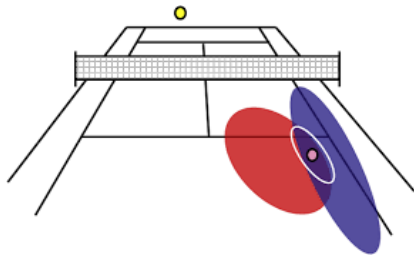
Let $f(s) = s^2$. Then

$$\mathbb{E}_{s \sim \mathcal{N}(0,1)}[f(s)] = \int_{-\infty}^{\infty} s^2 p(s) ds.$$

It is well-known that for a standard normal distribution,

$$\mathbb{E}[s^2] = 1.$$

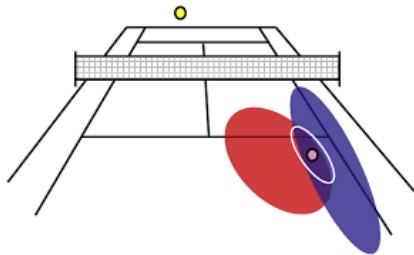
Perception as Inference



Determine the landing location

- Using sensory observations to determine the “state” of the world.
- Sensory information is usually incomplete or noisy so perception is inherently uncertain.

Perception as Inference



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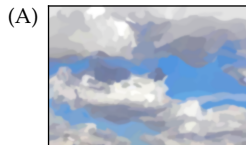
- Using sensory observations to determine the “state” of the world.
- Sensory information is usually incomplete or noisy so perception is inherently uncertain.
- A trained player has a generative model:

$$\mathbb{P}(\text{obs}|\text{state}) \in [0, 1]$$

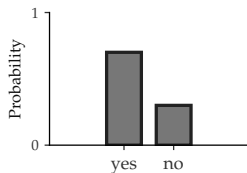
The Problem of Inference

- Given a new **observation**, determine which the **state** (e.g. the landing location).
- Inference is the process of going “in reverse” in Figure 1 to obtain $\mathbb{P}(\text{state}|\text{obs}) \in [0, 1]$
- Note on terminology: Never say “the likelihood of the observation.” The likelihood is always a *function of the hypothesized state of the world*. It is equal to the probability of a given **observation** under the hypothesized **state** of the world.

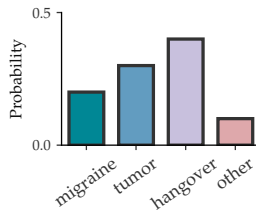
The Problem of Inference



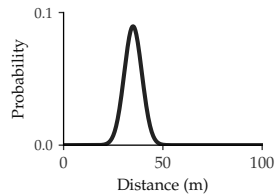
$P(\text{it is going to rain} \mid \text{cloudy sky})$



$P(\text{cause} \mid \text{symptoms})$



$P(\text{distance to car} \mid \text{visual image})$



Bayesian Approach

- We refer to states of the world as hypotheses H_i , $i \in \{1, \dots, N\}$ and $\frac{\mathbb{P}(H_i)}{\mathbb{P}(H_j)}$ as the *prior* odds.

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$$\underbrace{\frac{\mathbb{P}(H_i|\text{Obs})}{\mathbb{P}(H_j|\text{Obs})}}_{\text{posterior odds}} = \underbrace{\frac{\mathbb{P}(\text{Obs}|H_i)}{\mathbb{P}(\text{Obs}|H_j)}}_{\text{likelihood ratio}} \times \underbrace{\frac{\mathbb{P}(H_i)}{\mathbb{P}(H_j)}}_{\text{prior odds}}$$

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- This is equivalent to:

$$\frac{\mathbb{P}(H_i|\text{Obs})}{\mathbb{P}(\text{Obs}|H_i)\mathbb{P}(H_i)} = \frac{\mathbb{P}(H_j|\text{Obs})}{\mathbb{P}(\text{Obs}|H_j)\mathbb{P}(H_j)} = \kappa$$

for some constant $\kappa > 0$.

Why Equivalent?

$$\mathbb{P}(H_i \mid \text{Obs}) = \frac{\mathbb{P}(\text{Obs} \mid H_i) \mathbb{P}(H_i)}{\mathbb{P}(\text{Obs})}$$

$$\frac{\mathbb{P}(H_i \mid \text{Obs})}{\mathbb{P}(\text{Obs} \mid H_i) \mathbb{P}(H_i)} = \frac{1}{\mathbb{P}(\text{Obs})}$$

$$\kappa = \frac{1}{\mathbb{P}(\text{Obs})}$$

- This value does not depend on the hypothesis H_i or H_j .
- It is just a normalizing constant ensuring that all posterior probabilities sum to 1.

Bayesian Approach

- Hence, the *posterior* likelihood can be written as:

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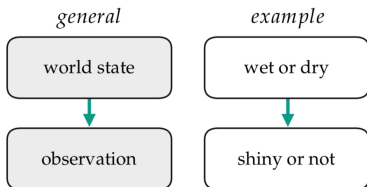
$$\kappa = \frac{1}{\sum_i \mathbb{P}(\text{Obs}|H_i)\mathbb{P}(H_i)}$$

- Hence in general,

$$\mathbb{P}(H_i|\text{Obs}) = \frac{\mathbb{P}(\text{Obs}|H_i)\mathbb{P}(H_i)}{\sum_i \mathbb{P}(\text{Obs}|H_i)\mathbb{P}(H_i)}$$

Bayesian Approach: Example 1

(A) Generative model



Distribution of world state

$$p(\text{wet}) = 0.1$$

$$p(\text{dry}) = 0.9$$

Distribution of observations

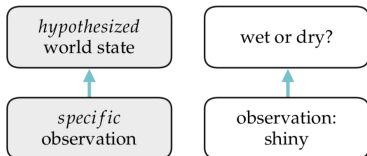
$$p(\text{shiny}|\text{wet}) = 0.8$$

$$p(\text{not shiny}|\text{wet}) = 0.2$$

$$p(\text{shiny}|\text{dry}) = 0.4$$

$$p(\text{not shiny}|\text{dry}) = 0.6$$

(B) Inference



Prior probabilities

$$\text{Prior (wet)} = p(\text{wet}) = 0.1$$

$$\text{Prior (dry)} = p(\text{dry}) = 0.9$$

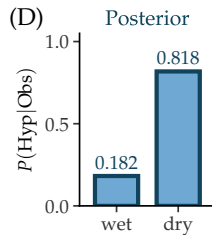
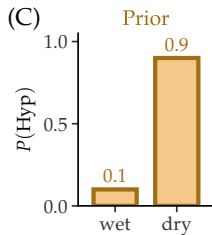
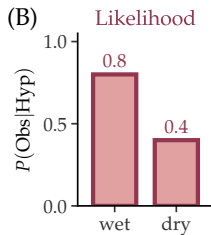
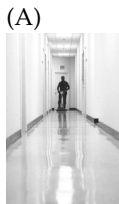
Likelihoods

$$\text{Likelihood (wet)} = p(\text{shiny}|\text{wet}) = 0.8$$

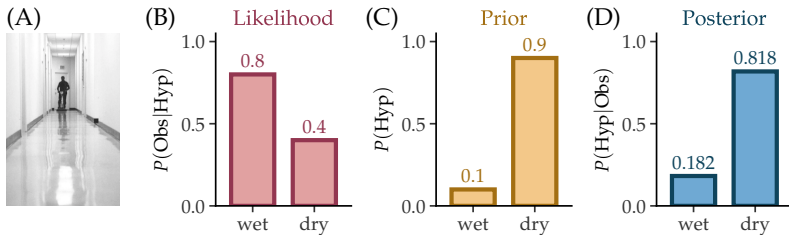
$$\text{Likelihood (dry)} = p(\text{shiny}|\text{dry}) = 0.4$$

- A generative model can be represented by a diagram with nodes and arrows.
 - Each node is a variable, with the observation(s) always at the bottom.
 - Each node is associated with a probability distribution.
 - Each node with an arrow pointing to it is associated with a conditional probability distribution.
 - Numbers are given for our shiny floor example.
- In inference, the observer has a specific observation (in the example: the floor is shiny) and tries to calculate the probabilities of the hypothesized world states.
 - The same numbers are now used as priors and likelihoods.
 - Inference “inverts” the generative model.

Bayesian Approach: Example 1



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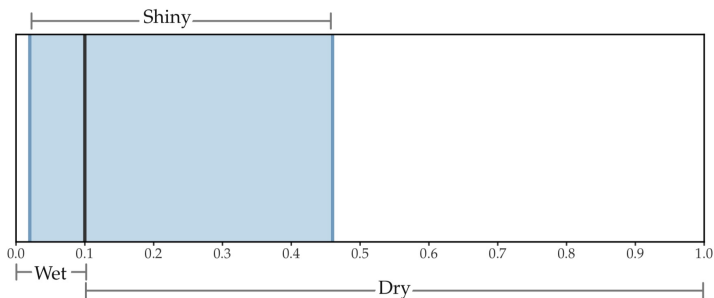


Hence, the posterior is

$$\mathbb{P}(\text{floor is wet} | \text{floor is shiny}) = \frac{0.8 \times 0.1}{0.8 \times 0.1 + 0.4 \times 0.9} = 0.182$$

$$\mathbb{P}(\text{floor is dry} | \text{floor is shiny}) = \frac{0.4 \times 0.9}{0.8 \times 0.1 + 0.4 \times 0.9} = 0.818$$

Bayesian Approach: Example 1



Areal representation of Bayesian inference:

- The posterior probability that the floor is wet, given that it is shiny, is the blue area within the left rectangle divided by the total blue area.
- The posterior probability that the floor is dry, given that it is shiny, is the blue area within the right rectangle divided by the total blue area.

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- The likelihoods $\mathbb{P}(Obs|H)$ and $\mathbb{P}(Obs|H^c)$ are estimated through empirical data.
- Let $\frac{\mathbb{P}(H)}{\mathbb{P}(H^c)}$ denote the *a priori* odds. Bayesian inference asserts that:

$$\frac{\mathbb{P}(H|Obs)}{\mathbb{P}(H^c|Obs)} = \frac{\mathbb{P}(Obs|H)}{\mathbb{P}(Obs|H^c)} \frac{\mathbb{P}(H)}{\mathbb{P}(H^c)}$$

Example 2: Covid Test

- For example, suppose

$$\mathbb{P}(\text{Obs} = \text{positive} | H = \text{sick}) = 0.95 \quad \mathbb{P}(\text{Obs} = \text{positive} | H = \text{not sick}) = 0.01$$

$$\text{and } \mathbb{P}(H = \text{sick}) = 0.1.$$

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$$\text{and } \mathbb{P}(H = \text{sick}) = 0.1.$$

- Then

$$\begin{aligned} \mathbb{P}(H = \text{sick} | \text{Obs} = \text{positive}) &= \frac{0.95 \times 0.1}{0.95 \times 0.1 + 0.01 \times 0.90} \\ &= 0.913 \end{aligned}$$

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- Hence,

$$P(H|Obs) \gg P(H^c|Obs)$$

Speech Recognition



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Communication is a fundamental human need: decoding for a 2-yo kid. Remarkably English speakers can usually understand each other even when accents differ. This is thanks to English built-in redundancy which allows "errors" to be tolerated. One might hear ...

... *"I need little little trail mix."*

Source: Essential Coding Theory. Guruswami et al. 2025, p.3.